

Problem 1

(a) Base Case:

For $n = 1$ it is trivial since $I \subseteq p_1$.

Inductive Step: Assume true for $2 \leq n - 1$. Consider n .

If I is a subset of any $n - 1$ unions of primes then we are done thus we can assume

$$I \not\subseteq \bigcup_{j \neq i} p_j$$

for all $i \in \{1, \dots, n\}$

Thus let

$$x_i \in I \setminus \bigcup_{j \neq i} p_j$$

If $x_i \in p_i$ since $I \subseteq p_1 \dots p_n$

Consider $x = x_1 + x_2 x_3 \dots x_n$

If $x \in p_i$ for $1 < i \leq n$ then $x - x_2 x_3 \dots x_n = x_1 \in p_i$ since $x_2 x_3 \dots x_n \in p_2 \cap p_3 \cap \dots \cap p_n$.

This is a contraction since $x_1 \notin p_2 \cup p_3 \dots \cup p_n$.

If $x \in p_1$ then $x - x_1 = x_2 x_3 \dots x_n \in p_1$ this is a contraction since p_1 is a prime ideal this means that some $x_i \in p_1$ for $i \neq 1$.

This $x \notin p_1 \cup p_2 \cup \dots \cup p_n$ which is a contraction thus

$$I \subset \bigcup_{j \neq i} p_j$$

for some $i \in \{1, \dots, n\}$ Thus by the Inductive Hypothesis we are done.

Problem 2

(a) Let $\varphi : A \rightarrow A[X]/(fX - 1)$ s.t. $\varphi(a) = [a]$.

Consider $\varphi(f^k)$ for $k \in \mathbb{N} \cup \{0\}$.

$$\varphi(f^k) = \varphi(f)^k = [f]^k.$$

Consider $[f]^k X^k = ([f]X)^k = 1$ since $fX - 1 = 0 \implies fX = 1$.

Thus, f^k is a unit.

Let $g(x) \in A[X]/(fX - 1)$. Thus $g(x) = a_0 + a_1X + \cdots + a_nX^n$ for $a_i \in A$

since $X = \varphi(f)^{-1}$ we get that

$$g(X) = a_0 + a_1 + f^{-1} + \cdots + a_n f^{-n}$$

Now combining terms we get

$$g(X) = \frac{a_0 f^n + a_1 f^{n-1} + \cdots + a_n}{f^n}$$

Thus

$$g(X) = \varphi(a')\varphi(f^n)^{-1}$$

Thus every thing is written in the form $\varphi(a)\varphi(s)^{-1}$ for $s \in \{1, f, f^2, \dots\}$

Let $b \in \ker \varphi$ thus $\varphi(b) = (fX - 1)g(X)$ for some $g(X) = g_0 + g_1X + \cdots + g_nX^n \in A[X]$

$$b = -g_0 + (fg_0 - g_1)X + (fg_1 - g_2)X^2 + \cdots + (fg_{n-1} - g_n)X^n + fg_nX^{n+1}$$

- (Constant)

$$g_0 = -b$$

- (X) $fg_0 - g_1 = 0$

$$g_1 = fg_0 = -fb$$

- $(X^i) fg_{i-1} - g_i = 0$

$$g_i = fg_{i-1} = f^i g_0 = -f^i b$$

- $(X^{n+1}) fg_n = 0$

$$fg_n = -f^{n+1}b = 0$$

Thus

$$f^{n+1}b = 0$$

Thus $\ker \varphi \subseteq \{a \in A \mid \exists k \geq 0, f^k[a] = 0\}$

Let $b \in \{a \in A \mid \exists k \geq 0, f^k a = 0\}$ thus $f^k b = 0$ for $k \geq 0$

Consider $-b(1 + fX + f^2X^2 + \cdots + f)(fX - 1)$

$$-b(1 + fX + f^2X^2 + \cdots + f^{k-1}X^{k-1})(fX - 1) = -b((fX)^k - 1)$$

finite geometric series

$$= b - bf^kX^k = b$$

Thus $b \in (fX - 1)$ thus $\varphi(b) = 0$ thus $b \in \ker \varphi$.

Thus, there is a unique isomorphism between A_f and $A[X]/(fX - 1)$

Therefore,

$$\mathbb{Z}[X]/(2X - 1) \cong \{1, 2, 4, \dots\}^{-1}\mathbb{Z} = A$$

Let $\varphi : A \rightarrow \mathbb{Q}$ s.t. $\varphi(a/2k) = a/2k$

Injective: This is clearly injective since $a/2k = b/2m$ in $\mathbb{Q}$ then $a2m = b2k \implies 1(a2m - b2k) = 0$ which satisfies the equivalence relation in A .

Surjective:

Let $a/b \in \mathbb{Q}$ then $a/b = 2a/2b$. Since $2b \in \{1, 2, 4, \dots\}$, we get that $2a/2b \in A$. Thus $\varphi(2a/2b) = a/b$.

Homomorphism:

$$\varphi(a/b)\varphi(c/d) = (a/b)(c/d) = ac/bd = ac/bd = \varphi(ac/bd)$$

$$\varphi(a/b) + \varphi(c/d) = (ad + cb)/bd = \varphi((ad + cb)/bd) = \varphi(a/b + c/d)$$

$$\varphi(1/1) = 1/1$$

Thus $A \cong \mathbb{Q}$

$$(\mathbb{Z}/6\mathbb{Z})[X]/(2X - 1) \cong \{1, 2, 4\}^{-1}(\mathbb{Z}/6\mathbb{Z}) \cong \mathbb{Z}/3\mathbb{Z}$$

We showed the last part in class

(b) Let φ be the natural homomorphism from $A \rightarrow S^{-1}A$

Injective: Let $a \in \ker(\varphi)$. Thus, $\varphi(a) = 0$. Thus, by definition of $S^{-1}A$ there is a $u \in S$ s.t. $ua = 0$ since $u \in S$ it is a unit thus we can multiply its inverse on both sides and get $a = 0$.

Thus injective

Surjective: The map is by definition surjective.

Homomorphism:

Proven in class it is a homomorphism

Thus $A \cong S^{-1}A$

If K is a field then $S^{-1}K$ where S is all units is actually just $(K \setminus \{0\})^{-1}K$ since all things in K are units except zero. Thus, by definition this is the same as $\text{Frac}(K)$ thus by the proof above $\text{Frac}(K) \cong K$

Consider $\text{Frac}(R) = (R \setminus \{0\})^{-1}R$

- We get Associativity, Commutativity, Additive/Multiplicative Inverse, Additive inverses, Distributivity for free.
- (Multiplicative Inverses) Let $x \in \text{Frac}(R)$ s.t. $x \neq 0$. Thus, $x = \varphi(a)\varphi(b)^{-1}$ for $a, b \in R \setminus \{0\}$
Consider $y = \varphi(b)\varphi(a)^{-1}$ this is well-defined since $a \neq 0$.
Thus $xy = \varphi(a)\varphi(a)^{-1}\varphi(b)\varphi(b)^{-1} = \varphi(1)\varphi(1)^{-1} = 1$.
- ($0 \neq 1$) Since two non zero things in R never multiply to zero we get that there is no element $s \in R \setminus \{0\}$ s.t. $s(1 - 0) = 0$.

Thus, $\text{Frac}(R)$ is a field thus $\text{Frac}(\text{Frac}(R)) = \text{Frac}(R)$

(e) Separate Proof: Show $S^{-1}R \cong S'^{-1}R$ where $S' = (S \setminus \text{units of } S) \cup \{1\}$ for non zero rings.

Assume that S' has more than just 1 since if it has just one then S is all units, so then you can look at (b).

- Show S' is multiplicative set with 1

It has 1 by definition.

Let $a, b \in S' \setminus 1$ then since $a, b \in S \implies ab \in S$

Assume ab is a unit in R with inverse $c \in R$ this means that

$$(ab)c = 1 \implies a(bc) = 1$$

Thus a is a unit which is a contradiction. Thus, $ab \in S'$

Also multiplicative with 1 is trivial.

Let $r/s \in S^{-1}R$

If $s \in S'$ then there is nothing to do.

If s is a unit, then choose any element $t \in S' \setminus \{1\}$ thus $st \in S$.

Assume st is also a unit then $(st)c = 1 \implies s(tc) = 1 \implies t = s^{-1}c^{-1}$ this t is a unit which is a contraction.

Thus $st \in S'$ now we can represent $r/s = rt/st$

Thus every element in $S^{-1}R$ can be written in the form of r/s' for $s' \in S'$

Now using the classifications as above take the natural map $\varphi : (S'^{-1})R \rightarrow S^{-1}R$ s.t. $\varphi(r/s') = r/s'$.

This is well defined for the reasons above.

This is a homomorphism since we're using the same multiplication and addition rules.

This is surjective for the reasons above.

Injective:

Assume consider $r/s', r'/s'' \in S^{-1}R$ s.t. $\varphi(r/s') = \varphi(r'/s'')$ thus $r/s' = r'/s''$ in $S^{-1}R$ thus there is an element in $t \in S$ s.t.

$$t(rs'' - r's') = 0$$

If $t \in S'$ then we are done.

If t is not in that set then we know it is a unit thus we can get $rs'' - r's' = 0$ since 1 is in S' we know thus means that these elements are equivalent in $S'^{-1}R$

Thus injective.

Thus $S^{-1}R \cong S'^{-1}R$

Now for $\mathbb{Z}/60\mathbb{Z}$ using the proof above all the units are things that are co prime with 60. Thus, we only have to look at things that divide 60, which are in the form $2^a 3^b 5^c$ where $a, b, c \geq 0$.

However, for a localization with $2^a 3^b 5^c$ for $a, b, c \geq 1$. This gives us an inverse of 2, 3, 5. Since $1/(2^a 3^b 5^c) \times 2^a 3^b 5^{c-1}/1 \equiv 1/5$. Thus this problem actually reduces to just choosing subsets of $\{2, 3, 5\}$ in your set S .

Thus

1. ($S = \{1\}$) This just gives us R by part above.

2. $(2 \in S)$ Just 2 in our set. Which is $\mathbb{Z}/15\mathbb{Z}$
3. $(3 \in S)$ Just 3 in our set. Which is $\mathbb{Z}/20\mathbb{Z}$
4. $(5 \in S)$ Just 5 in our set. Which is $\mathbb{Z}/12\mathbb{Z}$
5. $(2,3 \in S)$ Just 2,3 in our set. Which is $\mathbb{Z}/5\mathbb{Z}$
6. $(2,5 \in S)$ Just 2,5 in our set. Which is $\mathbb{Z}/3\mathbb{Z}$
7. $(3,5 \in S)$ Just 3,5 in our set. Which is $\mathbb{Z}/4\mathbb{Z}$
8. $(2,3,5 \in S)$ This gives us that everything in our set has an inverse, so for example $2 * 30 = 0 \implies 1 = 0$ since 2,30 have an inverse. Thus, we have the zero ring in this case.

(f) Let P be a prime ideal in $\mathbb{Z}[X]$. Since $P \cap \mathbb{Z}$ is a prime ideal in $\mathbb{Z}$ we get that $P \cap \mathbb{Z} = (0), (p)$ for p prime.

If $\mathbb{Z} \cap P = (0)$, then no integer is in P . Thus consider $\text{Frac}(\mathbb{Z})[X]$ since P does not meet $\mathbb{Z} \cap \{0\}$ this means its extension is a prime ideal in $\text{Frac}(\mathbb{Z})[X] = \mathbb{Q}[X]$

Since $\mathbb{Q}$ is a field we know that $\mathbb{Q}[X]$ is a *PID* thus $P^e = (f(X))$ for some irreducible polynomial in $\mathbb{Q}[X]$. Now let $g(X) \in \mathbb{Z}[X]$ s.t. $g(X)/c = f(X)$ for $c \in \mathbb{Z}$. Thus $(g(X)) = (f(X))$, also we can cancel the coefficients, so we get that their gcd is 1.

Separate:

Let $f(X) \in P$ then $f(X)/1 \in P^e \cap \mathbb{Z}[X]$

Let $f(X) \in P^e \cap \mathbb{Z}[X] \implies f(X) = h(X)/s$ for $h(X) \in \mathbb{Z}[X]$ and $s \in \mathbb{Z} \setminus \{0\} \implies sf(X) - h(X) = 0 \implies sf(X) = h(X) \in P$ since $s \notin P \implies f(X) \in P$ thus $P^e \cap \mathbb{Z}[X] = P$.

Consider $(g(X)) \subseteq \mathbb{Z}[X]$. Let $h(X) \in (g(X))$ thus $h(X) = g(X)q(X)$ for $q(X) \in \mathbb{Z}[X]$.

Now we can extend this to $\mathbb{Q}[X]$ just by dividing by 1, and we get $h(X) = g(X)q(X) = cf(X)q(X) \in \mathbb{Q}[X] \implies h(X) \in P^e \cap \mathbb{Z}[X] = P$

Let $h(X) \in P$ then extend this to $\mathbb{Q}[X]$ thus $h(X) \in P^e = (f(X)) = (g(X))$ thus $h(X) = g(X)q(X)$ for $q(X) \in \mathbb{Q}[X]$. Now get rid of all the denominators in $q(X)$ and we get that $zh(X) = g(X)q_1(X)$ for $q_1(X) \in \mathbb{Z}[X]$. Thus $zh(X) \in P^e \cap \mathbb{Z}[X] = P$, since $P \cap \mathbb{Z} = 0 \implies h(X) \in P$.

Thus, $P = (g(X))$ which is an irreducible polynomial in $\mathbb{Z}[X]$.

If $\mathbb{Z} \cap P = (p)$. Then consider the projection from $\mathbb{Z}[X] \rightarrow \mathbb{F}_p[X]$. Since there is a bijection between ideals in $\mathbb{Z}[X]$ that contain (p) and ideals in $\mathbb{F}_p[X]$. Thus $P/(p)$ is a prime ideal in $\mathbb{F}_p[X]$.

Since $\mathbb{F}_p$ is a field we know that the ideals are principle in $\mathbb{Z}[X]$ thus the prime ideals are (0) and $(f(X))$ for irreducible in $\mathbb{F}_p$

If $P \not\equiv (p) = (0)$ then $P = (p)$.

If $P \not\equiv (p) \neq (p)$ then it will equal $P \not\equiv (p) = (f(X))$ where $f(X)$ is an irreducible polynomial mod p . Thus, $P = (p, f(X))$

(i) Ring

For two germs $[f]$ and $[g]$ in $\mathbb{R}_x$, define:

$$[f] + [g] := [f + g] \quad \text{and} \quad [f] \cdot [g] := [f \cdot g].$$

Here, $f + g$ and $f \cdot g$ are defined on the intersection of the domains of definition of f and g .

Well-Defined: Suppose $[f] = [f']$ and $[g] = [g']$, meaning there exist open neighborhoods W_1 and W_2 of x such that

$$f|_{W_1} = f'|_{W_1} \quad \text{and} \quad g|_{W_2} = g'|_{W_2}.$$

Then on $W = W_1 \cap W_2$, we have

$$(f + g)|_W = f|_W + g|_W = f'|_W + g'|_W = (f' + g')|_W,$$

and similarly for the product. Thus, the operations are independent of the choice of representatives, making $\mathbb{R}_x$ a ring with additive identity $[0]$ and multiplicative identity $[1]$.

Local Ring

Let

$$\mathfrak{m}_x = \{[f] \in \mathbb{R}_x \mid f(x) = 0\}.$$

If $[f], [g] \in \mathfrak{m}_x$, then

$$(f + g)(x) = f(x) + g(x) = 0,$$

so $[f] + [g] \in \mathfrak{m}_x$. Also, for any $[h] \in \mathbb{R}_x$,

$$(f \cdot h)(x) = f(x)h(x) = 0,$$

so $[f] \cdot [h] \in \mathfrak{m}_x$.

Let $[f] \in \mathbb{R}_x$ with $[f] \notin \mathfrak{m}_x$. Then $f(x) \neq 0$. By the continuity of f , there exists an open

neighborhood U of x where $f(y) \neq 0$ for all $y \in U$. Define

$$g(y) = \frac{1}{f(y)} \quad \text{for } y \in U.$$

Since g is smooth on U , we have

$$[f] \cdot [g] = [f \cdot g] = [1].$$

Thus, $[f]$ is invertible.

Since every element not in $\mathfrak{m}_x$ is a unit, the complement of $\mathfrak{m}_x$ consists entirely of units, and hence $\mathfrak{m}_x$ is the unique maximal ideal of $\mathbb{R}_x$. Therefore, $\mathbb{R}_x$ is a local ring.

- (j) Let $x \in \mathfrak{N}(A)$ thus $x^n = 0$ for $n \in \mathbb{N}$ but since $0 \in p$ for all $p \in \text{Spec}(A)$ this implies by the definition of a prime ideal that $x \in p$ since for $x^n = 0 \in p$ some factor of x^n has to be in p but all the factors are x thus $x \in p$ for all $p \in \text{Spec}(A)$

Let $x \in p$ for all $p \in \text{Spec}(A)$. Assume for contraction that x is not nilpotent thus consider A_x . Since 0 was not in our multiplicative set we know that A_x is not the zero ring thus there exists a maximal ideal m thus prime. However this is a contradiction since there is a bijection from primes ideals in A_x to ideals prime ideals of A that do not meet $\{1, x, x^2, \dots\}$. Since $x \in p$ for all $p \in \text{Spec}(A)$. This means that x must be nilpotent.

Therefore, the intersection of all prime ideals is the nilradical.

Let I be an ideal in A . Let P be the intersection of all prime ideals of A containing I .

Let $x \in r(I)$ thus $x^n \in I$ for $n \in \mathbb{N}$. Thus, $x \in P$ by the same reason as above.

Let $x \notin r(I)$ thus no power of x is in I . Consider A/I s.t. the image of x is y consider the set $S = \{1, y, y^2, \dots\}$. Since $x \notin r(I)$ this means that y will not be nilpotent. Thus $(A/I)_y$ is non zero thus we can get a maximal ideal m which is prime. Thus, this corresponds to a prime ideal in A/I that does not meet S . Now contract this prime in A/I to A and we get a prime ideal in A that contains I but not x . Since if it did have x then we can take its image then the prime ideal in A/I would have y which is a contraction of it not having S . Thus, $x \notin P$.

Therefore $r(I) = P$

- (k) If $A = (0)$ then $\text{Spec}(0) = \emptyset$, so it is Hausdorff.

Assume $\text{Spec}(A)$ where $A \neq (0)$ is Hausdorff. Assume for contradiction there is a $p \in \text{Spec}(A)$ that is not maximal. Thus, there is a maximal ideal q s.t. $p \subsetneq q$. Thus,

there are open sets $p \in U$ and $q \in V$ s.t. $U \cap V = \emptyset$. Using the basis for this space we get that there is an $f \in A$ s.t. $q \in X_f \subseteq V$.

Since $p \subseteq q \implies f \notin p$. Thus, $p \in X_f$ which is a contradiction that $U \cap V = \emptyset$. Thus all prime ideals are maximal.

Assume every prime ideal of A is maximal. Let $p, q \in \text{Spec}(A)$. Since p, q are maximal you can find elements $f \in p$ and $f \notin q$.

Consider $S = A \setminus q$ for $S^{-1}A$. Since all prime ideals in $S^{-1}A$ are prime ideals in A that do not meet S we get that q^e is the unique prime ideal. Thus, for $g/1 \in q^e = r(0)$ by the last parts. Thus, $g^n/1 = 0/1$. Thus, there is an $a \in S$ s.t. $ag^n = 0$. Since $0 \in p \cap q$ and $g^n \notin p$ thus $a \in p$. Thus $p \in X_a$ and $p \notin X_f = X_{f_n}$ same for q .

By (a) we know that $X_{f_n} \cap X_a = X_{f_n a} = X_0 = \emptyset$. Thus, $\text{Spec}(A)$ is Hausdorff.

Problem 3

- (b) Assume f is a unit. Consider $X_f = \{p \in X \mid f \notin p\}$. Let $fg = 1$. If $f \in p \implies fg = 1 \in p$, but since p is proper ideal this means $f \notin p$. Thus $X_f = X$

Assume $X = X_f$. Thus, $f \notin p$ for all $p \in X$. Thus, f is not in any maximal ideal, since all maximal ideals are prime. Consider (f) . Assume for contraction it is a proper ideal. By Zorn's Lemma we know that $(f) \subseteq (m)$ where (m) is maximal but f is not in any maximal ideal thus $(f) = A$. Thus $fa = 1$ for $a \in A \implies f$ is a unit.

- (c) Assume f is nilpotent. Thus $f^k = 0$ for $k \in \mathbb{N}$. Since $0 \in p$ for all $p \in X$. This means that $f^k \in p$. By definition of a prime ideal the only way for $f^k \in p$ is if one of its factors is in p , but since all of its factors are f this means $f \in p$.

Assume $X_f = \emptyset$ Thus f is in the intersection of all the prime ideals which shown in problem 2 is the nilradical. Thus, f is nilpotent.

- (e) Proven in last homework that those basic open sets form a basis thus we only have to consider open covers of the form $\{X_{f_i}\}_{i \in I}$ since every open set can be written as a union of basis sets.

So let $\{X_{f_i}\}_{i \in I}$ be an open cover for X .

Thus,

$$X = \bigcup_{i \in I} X_{f_i} = \bigcup_{i \in I} (X \setminus V(f_i)) = X \setminus \bigcap_{i \in I} V(f_i)$$

Shown in last homework this becomes

$$X = X \setminus \bigcap_{i \in I} V(f_i) = X \setminus V\left(\bigcup_{i \in I} \{f_i\}\right) = X \setminus V((f_i \mid i \in I))$$

Since $X = X \setminus V((f_i \mid i \in I))$. This means that $V((f_i \mid i \in I)) = \emptyset$. If the ideal $(f_i \mid i \in I)$ is proper, then by Zorn's lemma it is contained in a maximal ideal m , but maximal ideals are prime so $m \in V((f_i \mid i \in I))$ which is a contraction thus $(f_i \mid i \in I) = A$.

This means there exists a finite combination of f_i and $a_i \in A$ s.t. $1 = a_1 f_{i_1} + a_2 f_{i_2} + \dots + a_n f_{i_n}$

Let $p \in X$. Suppose $p \notin X_{f_{i_1}} \cup \dots \cup X_{f_{i_n}}$

Thus, $f_{i_j} \in p$ for all $j \in \{1, \dots, n\}$ thus $a_1 f_{i_1} + a_2 f_{i_2} + \dots + a_n f_{i_n} = 1 \in p$ which is a contraction thus $p \in X_{f_{i_1}} \cup \dots \cup X_{f_{i_n}}$

Thus $X \subseteq X_{f_{i_1}} \cup \dots \cup X_{f_{i_n}}$

Therefore, $X = X_{f_{i_1}} \cup \dots \cup X_{f_{i_n}}$ thus X is quasi compact.